

ORIGINAL ARTICLE

WILEY

Hospitalization rate of respiratory syncytial virus-associated acute lower respiratory infection among young children in Suzhou, China, 2010–2014

Shaolong Ren¹  | Ting Shi² | Wei Shan¹ | Si Shen¹ | Qinghui Chen² |
Wanqing Zhang¹ | Zirui Dai¹ | Jian Xue² | Tao Zhang¹ | Jianmei Tian² |
Genming Zhao¹ 

¹Department of Epidemiology, School of Public Health, Fudan University, Key Laboratory of Public Health Safety, Ministry of Education, Shanghai, China

²Soochow University Affiliated Children's Hospital, Suzhou, China

Correspondence

Genming Zhao, PhD, Department of Epidemiology, School of Public Health, Fudan University, 130 Dong'an Road, Shanghai 200032, China.
Email: gmzhao@shmu.edu.cn

Funding information

the Investigator-Initiated Studies Program of Merck Sharp & Dohme Corp Grant, Grant/Award Number: 61457; Shanghai New Three-year Action Plan for Public Health, Grant/Award Number: GWV-10.1-XK16

Abstract

Background: There is a limited amount of data in China on the disease burden of respiratory syncytial virus- (RSV) associated acute lower respiratory infection (ALRI) among young children. This study aimed to estimate the hospitalization rate of RSV-associated ALRI (RSV-ALRI) among children aged 0–59 months in Suzhou, China.

Methods: All cases from children hospitalized with ALRI who were aged 0–59 months in Suzhou University Affiliated Children's Hospital during January 2010 to December 2014 were retrospectively identified. Detailed diagnosis and treatment data were collected by reviewing each individual's medical chart. In accordance with the World Health Organization (WHO) influenza disease burden estimation, the hospitalization rate of RSV-ALRI among children aged 0–59 months in Suzhou, China, was then estimated.

Results: Out of the 28,209 ALRI cases, 19,317 (68.5%) were tested for RSV, of which the RSV positive proportion was 21.3% (4107/19,317). The average hospitalization rate of RSV-ALRI for children aged 0–59 months was 14 (95% confidence interval [CI]: 14–14)/1000 children years, and that for children aged 0–5, 6–11, 12–23, and 24–59 months were 70 (95% CI: 67–73), 31 (95% CI: 29–33), 11 (95% CI: 10–12), and 3 (95% CI: 3–3)/1000 children years, respectively.

Conclusion: A considerable degree of RSV-ALRI hospitalization exists among children aged 0–59 months, particularly in those under 1 year of age. Therefore, an effective monoclonal antibody or vaccine is urgently needed to address the substantial hospitalization burden of RSV infection.

KEYWORDS

acute lower respiratory infection, children, China, hospitalization rate, respiratory syncytial virus